

Successful Role of Smart Technology to Combat COVID-19

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Abstract—Wild and uncontrolled spread of the corona pandemic over several months throughout the world is a global issue. To combat this global issue advancement in information technology is also used along with medicine, biotechnology, and medical Instrumentation. Machine learning, artificial intelligence, and data science are providing a great contribution to the battle with COVID-19. By using such technologies there are high prospects to overcome with this pandemic and to lead a standard routine as it was before the pandemic. This paper analyses different technologies used in different conditions such as social distancing and prevention, quarantine and isolation, COVID detection and testing, patient treatment and care, and hospital management. This paper introduces transparent planning, technological methods, and digital procedures, with the latest smart tools in various sectors that help to defeat the intensity of coronavirus.

Keywords—Corona pandemic, Artificial intelligence, Machine learning, Data Science

I. TECHNOLOGY USAGE FOR CUSTOMARY PRACTICE

A. Social Distancing

The practice of maintaining physical distancing by keeping at least 6 feet distance from the people who are not from households to avoid direct contact with people in public places to minimize exposure and reduce the transmission of corona infection.

- *Satellite monitoring:*

To combat the COVID-19 pandemic along with medicine, biotechnology, and medical Instrumentation space technologies as satellites are also utilized. Satellites provide information regarding the global issue of a pandemic like social distancing polices and stay at home at the macro and micro levels. Using satellite images information concerned with cities such as highways, tourist destinations, industrial areas and shopping locations activities can be scrutinized. Even critical emergency services information can also access such as organizing quarantine services, medical facilities, and mortuaries that can be located using satellite images.

However, during the period of the corona pandemic crisis, a lot of innovations and advancements in technologies are taking place. Globally there are various satellites which are already available and implementing services need to be squeezed as per the requirement. Several navigation, remote sensing,

commutation, and meteorological satellites are in good use to handle the COVID-19 pandemic crisis [3].

- *Facial recognition-powered CCTV cameras:*

Surveillance of public activities and their facial recognition is done by using CCTV in many areas around the world. Technology enterprise with neural networks with machine learning models loosely on the human brain to identify faces by providing millions of images. Sometimes AI Software is used to scan and create a profile for every citizen crossing in front of the lens and cameras and then compares them to the citizens in the surveillance list. This can help the monitoring of passengers returning from abroad, after landing they can be stopped for testing and quarantine to minimize the spread of the deadly virus. The CCTV surveillance system can also be used for contact tracing, tracking of an infected person's through his/her movements over a period of time.

B. Front End Warriors

Since the start of a pandemic, the generous selfless frontend warriors like health-care administrators, doctors, consultants, nurses, medicos, medical domestic staff, pathologists, emergency medical services, and ambulance drivers are serving people sacrificing and cutting off from their loved one and families. The Corona pandemic taught many lessons one among is the use of technology and infrastructure in a more efficient and better way in the field of the medical fraternity and by medical professionals.

- *Face Mask Detection:*

Using neural networks and machine learning models face mask detection can complete a search in under few seconds and even when people are wearing face masks by identifying them by their eye line. This helps in identifying and alerting people without a face mask to prevent the spread of the infectious coronavirus. Drones are also used for facial recognition and to announce warnings to the citizens do not step out in public places and workplaces without wearing face masks [13].

- *Autonomous Vehicles:*

These types of autonomous vehicles are helping in the delivery of essential goods like food items and medicines to hospitals and isolated patients. These vehicle mechanisms on the principal of self-driving start-up to fight the virus and to

protect healthcare authorities from the risk of contaminating viruses from people.

- *Open Access Data:*

Open access data helps citizen update educated regarding the corona virus and helps in taking precautions. Open access can facilitate research and information on COVID-19 to discover vaccination and to take public health measures to prevent or reduce the spread of the virus [14]. Data played a vital role in tracking and monitoring the spread of corona since the start of the pandemic. Information is collected as raw unprocessed data by using tools such as surveillance, CCTV, biometrics over a secure network that should be deployed for protecting information and secure transmission to ensure data is non-tempered [9]. Below table-1 defined are some of the major benchmarks where the open-access data are extremely used.

Table-1 major objectives of using open access data

Integral Role	Description
Research and Publications	Researchers can update and forward the research conducted on Corona to avoid the spread of the virus and to discover the vaccination.
Dashboard	Immediate accessibility of the daily update, and cumulative, averages, and comparative figures are defined.
Data visualization	Permits users to see the number of investigations conducted and the number of individuals who have tested negative, positive, or inconclusive locally and globally.
Hospital management	Helps in managing hospitals and clinics according to location.
Pharmaceuticals	It helps in providing medications in infected locations.
Contract tracing	Helps in tracking and tracing corona positive cases.
Surveillance	Helps in identifying people without precautions in public places to stop the spread of the virus.
Epidemic map service using GIS	Visualization and mapping of data on the globe which can help frequent travelers to track the corona status of a particular location.

- *Sanitization*

Drones: Drones are used for the Sanitization of Public Places, Hospitals & Tall buildings. Drones can be used to spray disinfectants on tall buildings. Drone operations are less expensive, easy to mobilize, faster, longer & safer than manual spraying by workers who can be exposed to disinfectant and COVID-19. Countries like China, India, Philippines, Indonesia, Colombia, UAE, and Chile have successfully used disinfecting drones to control the spread of coronavirus.

Electric street cleaning vehicles: Self-driving vehicles are operated as electric street cleaning vehicles; these vehicles are being used to disinfect hospitals and city streets.

- *COVID Detection and Testing*

Healthcare systems are tackling COVID-19 using digital technology to maintain core and critical clinical services. Virtual clinics are used as telemedicine consultations without physical crowding [2]. The use of AI-based chatbot helps recognize early symptoms in patients and start medication. The technologies which are highly productive for testing and detecting corona in hospitals and clinics are IoT devices integrated with AI, big data analytics, blockchain technology, and deep learning systems. By implementing these technologies in the health care system understanding of model risk associations, prediction outcomes and healthcare trends become more effective and efficient as shown in table-2.

Table- 2 Technology used for Monitoring, surveillance, detection, and prevention of COVID-19

Digital Technology	Monitoring, surveillance, detection, and prevention of COVID-19		Mitigation of impact	
	Example	Level	Example	Level
IoT	1. Simultaneous tracking and Simultaneous online Databases updating 2. Live to update of risk locality	High	1. Virtual clinics Public 2. information dissemination	High
Big data	1. Modelling of disease activity, growth, and regions of spread 2. Modelling of the awareness and susceptibility of countries struggling with a disease outbreak	High	1. Business modeling on medicinal supplies 2. Modelling of the utility of operating theatres and clinics with manpower projections	Moderate
AI	1. Detection of COVID-19 from chest scan imaging 2. Prediction of disease progression via imaging, clinical data, and AI	Moderate	1. AI can diagnose medical conditions unrelated to Corona 2. Health 'chatbots' for fulfilling people queries on Corona	High
Blockchain	1. Production and supply of COVID-19 vaccines when they are discovered and available 2. privileges such as Insurance to the patients suffering from Corona illness and demise	Low	Delivery of regular medicine from the local pharmacy to patients' home	Moderate

C. Mobile Applications

With the increase in the use of technology, mobile health applications (mHealth apps) are developed as software applications that are installed on smart devices such as smartphones and smart wearable's to broadcast health knowledge and resources [1]. mHealth app is easily adaptable, accessible, and acceptable for having the ability to support social distancing efforts, self-testing, quarantine monitoring, and contact tracing. mHealth provides solutions and strengthen facilities for rapid response teams, public health

mHealth App provides health services for individual citizens and the main objective is to provide health information and medical e-services to the users given by health organizations. E-services such as vital signs updates, testing, finding, isolating, and quarantining and providing treatment to every positive case of Corona, prescribing medicine, claiming medical leaves and promoting a healthy lifestyle, and also provide additional services associated to health awareness of individual citizens and their families. On the counterpart, mobile apps are complement but can never replace regular contact tracing methods. As not every person is having a smartphone and installed tracing apps, therefore use of mobile apps should be voluntary. Mobile technology offers an influential tool to simplify the defined task through applications that collect user data. The application uses Bluetooth and GPS techniques to connect via general approaches such as centralized contact tracing and decentralized contact tracing. These different applications are available on Google Play Store and App Store which works on a platform based on the web, Android, iOS platforms. List of countries with corona mobile application technology as given in the below table-3,

Table-3 List of countries with corona mobile application technology

S.No.	Countries	Mobile App used
1.	USA	CovidWatch
2.	Germany	The Corona DataSpende, Corona-Warn-App
3.	UK	COVID Symptom Tracker, NHS smartphone app
4.	Indonesia	PeduliLindungi
5.	Poland	Kwarantana Dommowa
6.	Paris	StopCovid App
7.	Singapore	TraceTogether App
8.	Australia	COVIDSafe app
9.	India	Aarogya Setu
10.	Saudi Arabia	Tawakkalna, Tabaud
11.	UAE	TraceCovid app
12.	Qatar	Ehteraz app
13.	Bahrain	BeAware Bahrain app
14.	Oman	Tarassud Plus mobile app
15.	Austria	Stopp Corona
16.	Azerbaijan	e-Tabib
17.	Bangladesh	Corona Tracer BD
18.	Brazil	Tô de Olho
19.	Canada	COVID Alert
20.	China	conjunction with Alipay and WeChat,
21.	Denmark	Smittestop
22.	France	StopCovid
23.	Ghana	GH Covid-19 Tracker App
24.	Hungary	VirusRadar
25.	Iceland	Rakning C-19
26.	Ireland	COVID Tracker app
27.	Italy	Immuni
28.	Jordan	AMAN App - Jordan
29.	New Zealand	NZ COVID Tracer
30.	Switzerland	SwissCovid

D. Smart Wearable

As the covid-19 pandemic intensifies, emerging technologies are implementing to ease the burden of health

care systems. Wearable devices are used to collect data about physical activity to track the general well-being of people such as body temperature, heart rate, sleep quality, and maintaining social distancing. Typically common smart wearable available is smart finger rings, smart bands, smartwatches, and smart clothing as well as smart eyeglasses [10].

Smart Watches: Smart Wearable watches with sensors constantly gather information about your heart rate, temperature, and blood oxygen could become personal warning systems for the COVID-19.

Smart Clothes: Smart clothes have built-in sensors for monitoring and tracking body temperature, ECG levels, stress levels, and sleep quality. They are comfortable and washable smart clothes that are available as underwear, shirts, and masks.

Respiratory Monitoring: To track and monitor the breathing pattern of patients suffering from coronavirus respiratory monitoring wearable devices are used. These wearable devices are used in both hospitals and the patient's homes. This helps in finding initial signs of patient health worsening and takes required measures [7].

MediBioSense – Real-Time Monitoring: These are the wearable solution as smart wearable patches to monitor and track patient's chronic diseases to protect from coronary heart disease, diabetes, and hypertension as they are highly prone to infect with the novel coronavirus.

E. Counseling For Physiological Stress Patients

During the COVID-19 pandemic, many people are suffering from stress and anxiety regarding their health and financial securities. In this situation, it is very important to maintain relations with both professional counselors and loved ones as citizens are never been so physically isolated [12]. Though there is no substitute for human existence there can be digital connectivity to be in touch with each other to help in relieving stress and practice self-care. As the Covid-19 pandemic is affecting globally its usual for the public to experience a wide range of emotions, feeling, and reactions such as, worry, feeling stressed Anxiety, fear, tearfulness, sadness, loss of interest in usually enjoyable activities, and due to this changes physical symptoms changes also occur such as increased heart rate, fatigue, stomach upset, frustration, anger, restlessness feeling helplessness, difficulty concentrating, sleeping disorder, feeling disconnected from others, and trouble relaxing. These drastic emotional changes are challenges that occurred due to loss of life, rapid changes in lifestyle such as study, work, and social gathering, travel restrictions, and social distancing and isolation to slow down the spread of the virus. Drastic changes in user behavior extents the researches in the field of human behavior and their stability [5].

The technology used to cope with stress, anxiety, or distress

- Get Your Mental Health Resources Digitally
- Digital consultation
- Virtual Dialogue through mobile apps
- Stay connected
- Seek accurate information to avoid misinformation

that can create panic and stress.

- Share positive Tip sheets and online resources
- Phone or online Counseling and Psychological Services (CAPS)
- Lief Therapeutics – Anxiety Reduction wearable smart device designed for monitoring the mental health of the user by tracking a user's heart rate variability (HRV).

II. TECHNOLOGY USAGE FOR PATIENTS TREATMENT/CARE

Advancement in technology is multiplying in an attempt to prevent contagion, effectively treat patients and simplify the pressure of healthcare workers, and also battling to develop new vaccines and optimize strategies for controlling the spread of the epidemic. Different technology has been used to monitor, detect, medical treatment delivery medicine and food to patients using as shown in the table-4.

Table-4 Technology Used for Patients Care both in-home and hospitals

Technology	Outlining Technology	Home Isolation	Hospital
Robotics	Robots are on the frontline to stop the spread of Covid-19. Robots are used to delivering food and medicine, diagnosis, transport medical samples, and conducting thermal imaging.	√	√
Autonomous Vehicles	Autonomous vehicles are used for delivering essential goods like food items and medicines.	√	√
Remote support in Covid-19 intensive care	This technology consist of headset helps surgeons to view near-eye display augmented reality by combining 3D-multispectral cameras, advanced computer analytics.	X	√
monocular AR headset	This smart-eye technology helps remote support for nurses and doctors in ICU during the corona crisis.	√	√
Photonic biosensors	Biosensor chip technology helps in food diagnostics, point-of-care applications, and environmental monitoring through precise and rapid analysis of various substances.	√	√
Breathing equipment	Breathing tools are a vital instrument to treat patients with advanced respiratory problems due to Corona. Patients are given oxygen through Continuous Positive Airway Pressure (CPAP) procedures that have avoided the need for invasive machine-driven ventilation.	√	√

III. GIS TECHNOLOGY USAGE FOR EPIDEMIC INFORMATION

Center for Systems Science and Engineering of Johns Hopkins University's has updated a real-time tracking map for following cases of COVID-19 throughout the globe, using the data collected from the World Health Organization (WHO), US Centers for Disease Control and Prevention (CDC), the Chinese Center for Disease Control and Prevention, and the European Center for Disease Prevention and Control.

A. Epidemic map service

Epidemic map service allows visualization and display of data related to COVID-19. This service presents geospatial positioning information and accurate data on the map regarding the status of infected areas such as villages, townships, cities, and countries. By this users can easily access the epidemic risk level of a particular location. This service can also be used for the distribution of COVID-19 cases in hospitals and to find an early warning, preventions, and control techniques [8].

B. Active updating system of pandemic prevention and control

This system is incorporated with multiple source data merged with epidemic prevention and control for administrative division maps. To achieve accurate positioning and management, the system supports the integration of protective data like key personnel data, isolation centers, quarantine centers, report epidemic data, and integrated medicinal treatment centers with spatial data like grid-centered map data and administrative divisions.

C. Pandemic condition awareness visual aid

This service visualizes the geographic scattering of new confirmed corona cases, accumulative corona diagnoses, and recovered corona cases through community grid maps and administrative divisions. It also provides the use of aggregation, heat, and scattered points to display the epidemic trend in a manner of spatiotemporal evolution, and displays the development trend of the epidemic in multiple dimensions.

D. Pandemic association research and decision visual aid

This service is used to locate transportation and commutation data such as bus, train, subway, and plane connected through the network based on geographic visualization and Big Data analytic through the real-time association with coordinated traffic epidemic observers and ticket information.

E. Mobile Pandemic chart

This provision allows people to observe pandemic information in real-time and conveniently on handy gadgets.

IV. TECHNOLOGY FOR DOCTORS/PHYSICIANS

A. Online Mode support in Covid-19 intensive care

This technology bids physician's smart head-worn augmented reality by conjoining 3D-multispectral cameras, near-eye spectacles, and innovative computer analytic. The tool comprises a head-set with an inbuilt camera that allows others to watch what you observe and offer guidance, also display facts or procedures [15]. Its workings are via wearable PC connected to wifi. The scholars have also established the other device, a monocular AR headset, consists of a camera that mechanism is with a smart-phone app. This smart-eye expertise could deal with distant online support for specialists and nurses in ICUs during the Corona calamity.

B. Photonic biosensors for coronavirus detection

It is mounting biosensor chip equipment that may make a way to major developments in point-of-care submissions, nutrition diagnostics, and ecological observing through the quick and accurate investigation of several elements [11]. It also improves coronavirus detection as it permits: real-time examination of many constituents and bodily factors; outcomes within few minutes; steadfast and safe data transmission, digital examination to escape prejudice; also instinctual and cost-effective procedure. Furthermore, the expertise is reduced, movable, and prepared for bulk manufacture. All these structures are higher to the detection methods now accessible [6].

C. On-the-fly detection

SARS-CoV-2 atoms are balls with 60 nm to 140 diameters. Imaging these with conservative visual systems is impossible owing to the 3-D determination boundary levied by deflection. However current advances in high imaging resolution enable determining visual facts outside this boundary, and developing ocular nanoscopy methods could be vital to combating the pandemic, as they may allow rapid and precise Covid-19 investigation [5].

D. Potential for tissue diagnostics

To understand the consequences of SARS-CoV-2 contagion on interior tissues is becoming critical. Countless roles of this virus are still unclear, and emergent indication displays that further, its recognized result on the lungs may also mutely disturb the brain, liver, gastrointestinal tract, and further body parts. The title standard for learning and detecting tissue reforms related to SARS-CoV-2 is excisional biopsy trailed by tissue dispensation and analysis. This method is gentle, invasive, costly, and focuses on sampling fault and informative inconsistency [16]. These boundaries frequently make this method unrealistic for healthcare systems around the sphere. Higher-harmonic generation (HHG) microscopy that attains enhanced tenacity and compassion also a difference in devices and deciding authority of Re-Scan HHG could considerably relieve, and in around other cases completely incredulous, the boundaries of traditional histopathology.

V. CONCLUSION

This paper discusses various ways globally managing to control the pandemic effectively which includes the significant role of the latest technologies which pay off their digital investment decisions in a most effective manner. Certainly, advanced technological involvement is vital, effective, and substantial in combating against Covid-19. Machine learning, artificial intelligence, and data science are providing a great contribution to the battle with COVID-19. AI's had an extreme contribution to a pandemic for effective management of resources, public awareness, carrying safety measures, and helping experts in enforcing strict regulations. Different digitally established sectors are discussed in the paper to get the proper view of their methods and working flow to achieve

a higher recovery rate without hampering much of their economy.

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